

标准答案及评分标准

2018年6月24日

一、填空题(每小题 4 分, 共 20 分)

1. $x + y - 1 = 0$

2. $-\frac{11}{3}$

3. $\int_0^1 dy \int_{2-y}^{1+\sqrt{1-y^2}} f(x, y) dx$

4. -2π

5. $|a| < e$

二、计算题(每小题 5 分, 共 20 分)

1. 解: 由 L , 视 x 为自变量, 有

$$\begin{cases} 4x + 6y \frac{dy}{dx} + 2z \frac{dz}{dx} = 0, \\ 6x + 2y \frac{dy}{dx} - 2z \frac{dz}{dx} = 0. \end{cases}$$

以 $(x, y, z) = (1, -1, 2)$ 代入并解出 $\frac{dy}{dx}, \frac{dz}{dx}$,

$$\text{得 } \frac{dy}{dx} = \frac{5}{4}, \frac{dz}{dx} = \frac{7}{8}, \quad \dots\dots\dots (3 \text{ 分})$$

$$\text{所以切线方程为 } \frac{x-1}{1} = \frac{y+1}{\frac{5}{4}} = \frac{z-2}{\frac{7}{8}},$$

$$\text{法平面方程为 } (x-1) + \frac{5}{4}(y+1) + \frac{7}{8}(z-2) = 0,$$

$$\text{即 } 8x + 10y + 7z - 12 = 0 \quad \dots\dots\dots (5 \text{ 分})$$

$$2. \text{ 解: } \frac{\partial z}{\partial x} = \left(f\left(\frac{y}{x}\right) - \frac{y}{x} f'\left(\frac{y}{x}\right) \right) + 2 f'\left(\frac{x}{y}\right)$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{y^2}{x^3} f''\left(\frac{y}{x}\right) + \frac{2}{y} f''\left(\frac{x}{y}\right) \quad \dots\dots\dots (3 \text{ 分})$$

$$\frac{\partial^2 z}{\partial x \partial y} = -\frac{y}{x^2} f''\left(\frac{y}{x}\right) - \frac{2x}{y^2} f''\left(\frac{x}{y}\right)$$

$$x \frac{\partial^2 z}{\partial x^2} + y \frac{\partial^2 z}{\partial x \partial y} = 0. \quad \dots\dots\dots (5 \text{ 分})$$

3. 解: 由题设, S 的方程为 $z = \sqrt{3(x^2 + y^2)}$, 因此

$$dS = \sqrt{1 + (z'_x)^2 + (z'_y)^2} dx dy = 2 dx dy$$

在 xoy 坐标面上的投影区域 $D_{xy}: x^2 + y^2 \leq 3$

$$I = \iint_S (x^2 + y^2) dS = 2 \iint_{D_{xy}} (x^2 + y^2) dx dy \quad \dots\dots\dots(3 \text{ 分})$$

$$\begin{aligned} &= 2 \int_0^{2\pi} d\theta \int_0^{\sqrt{3}} r^3 dr \\ &= 9\pi \quad \dots\dots\dots(5 \text{ 分}) \end{aligned}$$

4. 解:

$$\text{gradu} = \left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z} \right) = \left(\frac{x}{x^2 + y^2 + z^2}, \frac{y}{x^2 + y^2 + z^2}, \frac{z}{x^2 + y^2 + z^2} \right) \quad \dots\dots\dots(2 \text{ 分})$$

$$\begin{aligned} \text{div}(\text{gradu}) &= \text{div} \left(\frac{x}{x^2 + y^2 + z^2}, \frac{y}{x^2 + y^2 + z^2}, \frac{z}{x^2 + y^2 + z^2} \right) \\ &= \frac{\partial}{\partial x} \left(\frac{x}{x^2 + y^2 + z^2} \right) + \frac{\partial}{\partial y} \left(\frac{y}{x^2 + y^2 + z^2} \right) + \frac{\partial}{\partial z} \left(\frac{z}{x^2 + y^2 + z^2} \right) \\ &= \frac{y^2 + z^2 - x^2}{(x^2 + y^2 + z^2)^2} + \frac{x^2 + z^2 - y^2}{(x^2 + y^2 + z^2)^2} + \frac{x^2 + y^2 - z^2}{(x^2 + y^2 + z^2)^2} \\ &= \frac{1}{x^2 + y^2 + z^2}. \quad \dots\dots\dots(5 \text{ 分}) \end{aligned}$$

三、解: $\iint_D \left(\frac{t^2}{x} - 6y \right) f(x) dx dy = \int_0^t f(x) dx \int_0^x \left(\frac{t^2}{x} - 6y \right) dy = \int_0^t (t^2 - 3x^2) f(x) dx.$
 $\dots\dots\dots(3 \text{ 分})$

记 $F(t) = \int_0^t (t^2 - 3x^2) f(x) dx = t^2 \int_0^t f(x) dx - 3 \int_0^t x^2 f(x) dx$
 $F'(t) = 2t \int_0^t (f(x) - f(t)) dx \quad \dots\dots\dots(6 \text{ 分})$

由于 $f(x)$ 是单调减少的函数, 因此, $f(x) \geq f(t), 0 \leq x \leq t$,

故 $F'(t) \geq 0$, 所以 $F(t)$ 关于 t 单调增加. 因 $F(0) = 0$, 故对任意 $t \geq 0, F(t) \geq 0$,

亦即待证的不等式成立. $\dots\dots\dots(8 \text{ 分})$

四、解：设整个物体 Ω 的质心为 $(\bar{x}, \bar{y}, \bar{z})$, 由对称性得 $\bar{x}=0, \bar{y}=0$, 而 $\bar{z} = \frac{\iiint_{\Omega} \rho z dv}{\iiint_{\Omega} \rho dv}$,

根据题意, 只需: $\iiint_{\Omega} \rho z dv = \iiint_{\Omega} z dv = 0$ 即可. (2 分)

$$\begin{aligned} 0 &= \iiint_{\Omega} z dv = \int_{-h}^0 z dz \iint_{x^2+y^2 \leq 1} dx dy + \int_0^1 z dz \int_{x^2+y^2 \leq 1-z^2} dx dy \\ &= \pi \int_{-h}^0 z dz + \pi \int_0^1 z(1-z^2) dz \quad \dots\dots\dots (4 \text{ 分}) \\ &= -\frac{\pi}{2} h^2 + \frac{1}{4} \pi \end{aligned}$$

解得 $h = \frac{\sqrt{2}}{2}$ 时, 整个物体的质心恰好在半球的球心处. (6 分)

五、解：设所求平面方程为 $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1, a > 0, b > 0, c > 0$. 由题意得

$$\frac{2}{a} + \frac{1}{b} + \frac{1}{3c} = 1$$

题目转化为对函数 $V = \frac{1}{6}abc$

在约束条件 $\frac{2}{a} + \frac{1}{b} + \frac{1}{3c} = 1$ 下的最小值. 即为条件极值问题. (3 分)

构造函数: $F(a, b, c, \lambda) = \frac{1}{6}abc + \lambda \left(\frac{2}{a} + \frac{1}{b} + \frac{1}{3c} - 1 \right)$

$$\left\{ \begin{array}{l} F'_a = \frac{bc}{6} - \frac{2\lambda}{a^2} = 0 \\ F'_b = \frac{ac}{6} - \frac{\lambda}{b^2} = 0 \\ F'_c = \frac{ab}{6} - \frac{\lambda}{3c^2} = 0 \\ F'_\lambda = \frac{2}{a} + \frac{1}{b} + \frac{1}{3c} - 1 = 0 \end{array} \right. , \quad \dots\dots\dots (6 \text{ 分})$$

得唯一解 $a=6, b=3, c=1$, 所求平面为 $\frac{x}{6} + \frac{y}{3} + \frac{z}{1} = 1$ (8 分)

六、解：(1) 记 $X = 2xy$, $Y = Q(x, y)$ 由题意, 有

$$\frac{\partial X}{\partial y} = \frac{\partial Y}{\partial x}, \quad \text{即} \quad 2x = \frac{\partial Q(x, y)}{\partial x}, \quad Q(x, y) = x^2 + C(y) \quad \dots\dots\dots (2 \text{ 分})$$

$$\int_{(0,0)}^{(t,1)} 2xydx + Q(x, y)dy = t^2 + \int_0^1 C(y)dy$$

$$\int_{(0,0)}^{(1,t)} 2xydx + Q(x, y)dy = t + \int_0^t C(y)dy$$

$$\text{由条件得:} \quad t^2 + \int_0^1 C(y)dy = t + \int_0^t C(y)dy$$

$$\text{求导得} C(y) = 2y - 1, \quad Q(x, y) = x^2 + 2y - 1. \quad \dots\dots\dots (5 \text{ 分})$$

$$\begin{aligned} (2) \text{ 原函数 } u(x, y) &= \int_{(0,0)}^{(x,y)} 2xydx + (x^2 + 2y - 1)dy + C \\ &= \int_0^x 0dx + \int_0^y (x^2 + 2y - 1)dy + C \\ &= \int_0^x 0dx + \int_0^y (x^2 + 2y - 1)dy + C \\ &= x^2y + y^2 - y + C \end{aligned}$$

$$\text{所求原函数为 } u(x, y) = x^2y + y^2 - y + C \quad \dots\dots\dots (8 \text{ 分})$$

$$\text{七、解:} \quad \lim_{n \rightarrow \infty} \left| \frac{u_{n+1}(x)}{u_n(x)} \right| = \lim_{n \rightarrow \infty} \frac{2n-1}{2n+1} x^2 = x^2$$

$|x| < 1$ 时, $\sum_{n=1}^{\infty} u_n(x)$ 绝对收敛; $|x| > 1$ 时, $\sum_{n=1}^{\infty} u_n(x)$ 发散 (因为 $u_n(x)$ 不趋于 0).

$$\text{收敛半径 } R = 1, \quad \text{收敛域为 } [-1, 1]. \quad \dots\dots\dots (3 \text{ 分})$$

$$\text{设 } S(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n-1} x^{2n-1} (-1 \leq x \leq 1), \text{ 则}$$

$$S'(x) = \sum_{n=1}^{\infty} (-1)^{n-1} x^{2n-2} = \frac{1}{1+x^2}$$

$$\text{又因为} \quad S(x) = S(0) + \int_0^x \frac{1}{1+t^2} dt = \arctan x, \quad \dots\dots\dots (7 \text{ 分})$$

$$\text{所以} \quad \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n-1} x^{2n} = xS(x) = x \arctan x (x \in [-1, 1]) \quad \dots\dots\dots (8 \text{ 分})$$

八、解:

$$f(x) = \frac{1}{(x+3)(x+1)} = \frac{1}{2} \left(\frac{1}{x+1} - \frac{1}{x+3} \right)$$

$$= \frac{1}{2} \left[\frac{1}{2} \frac{1}{1 + \frac{x-1}{2}} - \frac{1}{4} \cdot \frac{1}{1 + \frac{x-1}{4}} \right] \quad \dots\dots\dots (2 \text{ 分})$$

$$= \frac{1}{2} \left[\sum_{n=0}^{\infty} (-1)^n \frac{(x-1)^n}{2^{n+1}} - \sum_{n=0}^{\infty} (-1)^n \frac{(x-1)^n}{4^{n+1}} \right]$$

$$= \sum_{n=0}^{\infty} (-1)^n \left(\frac{1}{2^{n+2}} - \frac{1}{2^{2n+3}} \right) (x-1)^n$$

其收敛域为: $-1 < x < 3$. \dots\dots\dots (6 \text{ 分})

$$a_{10} = \frac{1}{2^{12}} - \frac{1}{2^{23}} = \frac{f^{(10)}(1)}{10!},$$

$$f^{(10)}(1) = \left(\frac{1}{2^{12}} - \frac{1}{2^{23}} \right) 10!. \quad \dots\dots\dots (8 \text{ 分})$$

九、解: 添加辅助面 $S_0: z=0, x^2+y^2 \leq 4$, 取上侧;

$S_1: z=1, x^2+y^2 \leq 3$, 取下侧.

Ω 为 Σ 与 S_0, S_1 所围成的空间区域. \dots\dots\dots (2 \text{ 分})

$$I = \frac{1}{4} \oiint_{\Sigma+S_0+S_1} x^2 dydz + y^2 dzdx + z^2 dxdy - \frac{1}{4} \iint_{S_0} x^2 dydz + y^2 dzdx + z^2 dxdy$$

$$- \frac{1}{4} \oint_{S_1} x^2 dydz + y^2 dzdx + z^2 dxdy \quad \dots\dots\dots (4 \text{ 分})$$

$$\frac{1}{4} \oint_{\Sigma+S_0+S_1} x^2 dydz + y^2 dzdx + z^2 dxdy$$

$$= -\frac{1}{2} \iiint_{\Omega} (x+y+z) dv \quad (\text{利用高斯公式})$$

$$= 0 - \frac{1}{2} \iiint_{\Omega} z dv \quad (\text{利用对称性})$$

$$\begin{aligned}
&= -\frac{1}{2} \int_0^1 z dz \iint_{x^2+y^2 \leq 4-z^2} dx dy = -\frac{\pi}{2} \int_0^1 (4z - z^3) dz \\
&= -\frac{7}{8} \pi \quad \dots\dots\dots (6 \text{ 分})
\end{aligned}$$

$$\frac{1}{4} \oiint_{S_0} x^2 dy dz + y^2 dz dx + z^2 dx dy = 0$$

$$\frac{1}{4} \oiint_{S_1} x^2 dy dz + y^2 dz dx + z^2 dx dy = -\frac{1}{4} \iint_{x^2+y^2 \leq 3} dx dy = -\frac{3}{4} \pi$$

$$I = -\frac{7}{8} \pi - (0 - \frac{3}{4} \pi) = -\frac{1}{8} \pi. \quad \dots\dots\dots (8 \text{ 分})$$

十、 解: 因为 $\lim_{x \rightarrow 0} \frac{f(x)}{x} = 0$, 而 $f(x)$ 在 $x=0$ 的某邻域内有二阶连续导数, 所以

$$f(0) = \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{f(x)}{x} \cdot x = 0.$$

$$f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{f(x)}{x} = 0. \quad \dots\dots\dots (2 \text{ 分})$$

$f(x)$ 在 $x=0$ 处的麦克劳林公式为:

$$f(x) = f(0) + f'(0)x + \frac{1}{2} f''(\xi) \cdot x^2 = \frac{1}{2} f''(\xi) \cdot x^2 (\xi \in (0, x)).$$

$$\text{所以 } f\left(\frac{1}{n}\right) = \frac{1}{2} f''(\xi) \cdot \frac{1}{n^2}, \text{ 其中 } (\xi \in (0, \frac{1}{n})). \quad \dots\dots\dots (5 \text{ 分})$$

$$\text{由于 } \lim_{x \rightarrow 0} \frac{\left| f\left(\frac{1}{n}\right) \right|}{\frac{1}{n^2}} = \lim_{n \rightarrow \infty} \frac{1}{2} |f''(\xi)| = \frac{1}{2} |f''(0)|, \text{ 而级数 } \sum_{n=1}^{\infty} \frac{1}{n^2} \text{ 收敛,}$$

$$\text{所以级数 } \sum_{n=1}^{+\infty} f\left(\frac{1}{n}\right) \text{ 绝对收敛.} \quad \dots\dots\dots (6 \text{ 分})$$